Elouann Lucas

France · elouannlucas56@gmail.com · (+33) 640586765
elucas0.github.io/portfolio/ · linkedin.com/in/elouann-lucas/

Summary

Software engineer and recent graduate of the Erasmus Mundus Joint Master in Digital Earth (GeoData Science track), with a prior Bachelor's degree in Computer Science. Combines professional software engineering experience in full-stack development with expertise in Earth Observation data analysis and processing.

Work Experience

Visiting Researcher March 2026 – July 2026
ESA Φ-lab *Frascati, Italy*

- Researched on-board processing pipelines for the Φ-sat-2 satellite mission.
- Evaluated the robustness of geospatial foundation models against data drift by simulating physically-based image degradation for flood segmentation.
- Implemented and tested processing scripts for multi-band satellite data under Linux environments.

Application Developer – Work-Study September 2022 – August 2024
CGI France *Larmor-Plage, France*

- Developed backend and frontend features for an enterprise ERP platform.
- Designed microservices architecture to modularize and decouple a monolithic legacy system.
- Authored technical specifications and architectural documentation for the internal development framework.
- Participated in code integration, testing, and deployment workflows.

Education & Qualifications

Copernicus Master in Digital Earth (M.Sc.) September 2024 – September 2026
Université de Bretagne Sud / Paris Lodron Universität Salzburg *France / Austria*

- GeoData Science track: Big Data processing, Deep Learning, Machine Learning, and efficient remote sensing data pipelines.
- Main curriculum: Earth Observation missions (Sentinel, SAR, VHR optical), Copernicus Services, Copernicus OData API, OGC interface standards, and spatial databases.

Bachelor of Computer Science (B.Sc.) September 2021 – August 2024
Université de Bretagne Sud *Vannes, France*

- Specialization in Software and Applications Development (180 ECTS).
- Coursework: Software architecture, Linux system administration, DevOps (Docker, Jenkins, CI/CD), relational database management, and object-oriented programming.

Selected Projects

Scalable Satellite Data Pipeline on AWS Copernicus Master – GeoDSc
Code Repository

- Engineered a distributed data ingestion and pre-processing pipeline deployed on a multi-node AWS EMR cluster.

- Streamed and converted large-scale multi-spectral Sentinel-1 and Sentinel-2 data using PySpark and Petastorm.
- Handled end-to-end data format conversions for model training workflows.

Machine Learning Land Cover Classification Copernicus Master – GeoDSc
Code Repository

- Built automated Python processing workflows to extract features and classify land cover from VHR imagery.
- Implemented model validation and error analysis comparing Random Forest, SVM, and KNN classifiers.

Technical Skills

Programming & Scripting: Python, Java, TypeScript, Bash, SQL.
Software Engineering: Microservices Architecture, Technical Specifications, RESTful Concepts, Modular Design.
DevOps & Systems: Linux, Docker, CI/CD (Jenkins), Git, Cloud Infrastructure (AWS EMR).
Earth Observation & Data: Sentinel-1/2, Copernicus OData API, OGC Standards, BigEarthNet, PySpark, Petastorm, QGIS, Spatial Databases.
Languages: French (Native), English (C1).